

FAST TRANSIENT STABILITY ASSESSMENT REVISITED

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Abstract. Energy Liapunov-like criteria are devised, their essential characteristics are assessed, and, depending upon the particular needs they are called to meet, the most appropriate among them are identified. The criteria derive from the conjunction of various Liapunov-like functions ("global", "individual", and Taylor series based ones) with two different estimates of practical stability domains; bound to these criteria, rules for the proper identification of the system "relevant" machines are also proposed. Many variants are thus envisaged and investigated through a large number of simulations, carried out on eight different power systems. They allow highlighting criteria's salient merits, and suggesting appropriate selection among them.

1. INTRODUCTION

Liapunov's stability theorems provide sufficient but not necessary conditions, i.e. conservative stability assessment. Attempts to apply them to power system transient stability analysis confirm the conservative character with, moreover, uncertain degree of conservativeness. Indeed, as is well known, this degree varies significantly and randomly not only with the power systems but also, for a given power system with the kind and location of the considered disturbances, even when considering the same undisturbed (pre-fault and post-fault) configurations. Another well known shortcoming of Liapunov's criterion applied to power systems is the difficulty to construct practicable Liapunov functions for detailed system modelling.

For a fast appraisal of the system behaviour, the former question is even more crucial than the latter one. Indeed, while simplified modelling may be acceptable for a first and fast guess of the system tendency, alleviating the method's conservativeness is of paramount importance to make it workable. This paper aims at developing efficient methodologies appropriate to such a first and fast stability assessment. More explicitly, it aims at devising energy Liapunov-like direct criteria, at exploring their salient characteristics, and at proposing among them those which are the most likely to suitably meet particular practical requirements.

To this purpose, two families of energy type functions are at first envisaged, the "global" and the "individual" ones; within each family many variants are considered. Moreover, Taylor series based methods are developed; among them, those deriving from a two-term expansion exhibit a priori quite attractive analytical properties.

On the other hand, two approaches are considered for estimating practical stability domains, viz. the "Kyoto" and the "acceleration" ones.

Further, bound to the above functions and stability domains, the important question of "relevant" sys-

tem machines is examined and rules for their proper identification are set forth.

The combination of the different energy type functions with the stability domain estimates provides a large number of variants, eight of which are more extensively investigated in the paper. The investigations involve a large number of simulations, carried out on eight different real power networks and test systems, chosen so as to cover a wide range of system characteristics. On the other hand, two groups of severe disturbances are simulated, viz. standard three-phase short circuits, which generally impose the most stringent stability conditions, and load sheddings and line trippings which generally imply milder ones. The comparisons use the standard stability assessment, i.e. the system's capability to face a large disturbance in terms of its corresponding critical clearing time. Admittedly, the use of such a measure of a system's robustness from its transient stability viewpoint might be arguable. However, for comparison purposes it shows particularly convenient. By no means does it imply that the devised direct criteria are designed to solely comply with the requirements of this conventional stability analysis; rather, we believe that they are flexible enough to adapt to various planning and real time operation purposes.

N.B. The reader is presumed to be familiar with the subject. For additional information, he may refer to survey publications, such as Refs. [1 to 3].

2. ENERGY LIAPUNOV-LIKE FUNCTIONS :
FOUNDATIONS AND BUILDING BLOCKS

2.1. NOTATION

As a convenience to the reader we collect below some of the more frequently used symbols in several places. Other symbols are defined at the place they are first introduced.

n number of system's generators
 ω_0 synchronous speed of the system
 δ_i rotor angle of i -th generator relative to a reference rotating at the synchronous speed
 ω_i rotor speed of i -th generator above the synchronous speed : $\omega_i = \dot{\delta}_i$
 P_{mi} mechanical input to i -th generator from its turbine
 P_{ei} electrical output from the i -th generator
 M_i generator inertia coefficient
 E_i i -th generator's internal voltage
 Y admittance matrix reduced to the generators' internal nodes; let $[Y]_{ij}$ denote its ij -th term
 Y_{ij}, θ_{ij} modulus, argument of the term $[Y]_{ij}$
 $G_{ij} = Y_{ij} \cos \theta_{ij}$ conductance of the above term
 $B_{ij} = Y_{ij} \sin \theta_{ij}$ susceptance of the above term
 $\delta_{in} = \delta_i - \delta_n$; $\tilde{\delta}_i = \delta_i - \delta_0$; $\omega_{in} = \omega_i - \omega_n$; $\tilde{\omega}_i = \omega_i - \dot{\delta}_0$
 $P_i = P_{mi} - E_i^2 G_{ii}$; $P_{ij} = P_i M_j - P_j M_i$; $M = \sum_{i=1}^n M_i$
 $\delta_0 = M^{-1} \sum_{i=1}^n M_i \delta_i$
 t_{0+} time immediately following a disturbance inception
 γ_i acceleration of i -th generator at t_{0+}

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SEP Stable Equilibrium Point (solution) of the system in its post-fault configuration
 UEP Unstable Equilibrium Point surrounding SEP
 (δ, ω) state vector with variables δ_{in}, ω_{in} , $i=1,2,\dots,n-1$
 $(\tilde{\delta}, \tilde{\omega})$ state vector with variables $\tilde{\delta}_i, \tilde{\omega}_i$, $i=1,2,\dots,n$
 $(\delta^s, 0)$ value of the state vector at the SEP
 $(\delta^{ui}, 0)$ value of the state vector at the i -th UEP
 CCT Critical Clearing Time
 SDE (PSDE) Stability Domain Estimate (Practical SDE)
 TC Transfer Conductance(s)

2.2. POWER SYSTEM DESCRIPTION

In the standard simplified description of an n -machine system, the absolute motion of the i -th machine when all types of damping are neglected is expressed by

$$M_i \ddot{\delta}_i = P_{mi} - P_{ei} \quad (1)$$

where

$$P_{ei} = E_i^2 Y_{ii} \cos \theta_{ii} + \sum_{j=1, j \neq i}^n E_i E_j Y_{ij} \cos(\delta_i - \delta_j - \theta_{ij}) \quad (2)$$

Note that eqs. (1) and (2) are valid in either the fault-on and post-fault periods, depending upon the values of the parameters.

With the current simplifying assumptions (constant P_{mi} s and M_i s during the transients), the motion of the system taken as a whole has to be described with respect to a reference. A possible reference is one of the system generators. Choosing arbitrarily the n -th as such, we derive in the $2(n-1)$ -dimensional state space the equations of motion

$$\begin{aligned} \dot{\delta}_{in} &= \omega_{in} \\ \dot{\omega}_{in} &= (P_{mi} - P_{ei}) M_i^{-1} - (P_{mn} - P_{en}) M_n^{-1} \\ &= f_{in}(\delta_{jn}) \quad (i, j=1, 2, \dots, n-1) \end{aligned} \quad (3)$$

Another possible frame of reference is the so-called "Center of Angle" (COA) defined by [4 to 8] :

$$\delta_o = \frac{1}{M} \sum_{i=1}^n M_i \delta_i \quad (4)$$

Its motion is governed by

$$M \ddot{\delta}_o = \sum_{i=1}^n (P_{mi} - P_{ei}) = P_{COA} \quad (5)$$

The resulting equations are

$$\begin{aligned} \dot{\tilde{\delta}}_i &= \tilde{\omega}_i \\ \dot{\tilde{\omega}}_i &= P_{mi} - P_{ei} - \frac{M_i}{M} P_{COA} \quad i=1, 2, \dots, n \end{aligned} \quad (6)$$

Remark. From a theoretical viewpoint, it makes no difference using either of the above two frames of reference; similarly, the choice of the reference generator (n) is in principle indifferent.

2.3. ENERGY TYPE LIAPUNOV FUNCTIONS

2.3.1. The standard (global) Liapunov function and its approximate corrective terms

Rigorously, a Liapunov function may be constructed for state model (3) [respectively (6)] only when neglecting transfer conductances (TC), i.e. when setting $\theta_{ij} = \pi/2 \forall i, j$ ($i \neq j$). It expresses by

$$\begin{aligned} V(\delta, \omega) &= V_K(\omega) + V_P(\delta) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left\{ \frac{1}{2} M_i M_j \omega_{ij}^2 - P_{ij} (\delta_{ij} - \delta_{ij}^s) \right. \\ &\quad \left. - M E_i E_j B_{ij} (\cos \delta_{ij} - \cos \delta_{ij}^s) \right\} \end{aligned} \quad (7)$$

or respectively by

$$\begin{aligned} V(\tilde{\delta}, \tilde{\omega}) &= V_K(\tilde{\omega}) + V_P(\tilde{\delta}) = \sum_{i=1}^n \frac{1}{2} M_i \tilde{\omega}_i^2 - \sum_{i=1}^n P_i (\tilde{\delta}_i - \tilde{\delta}_i^s) \\ &\quad - \sum_{i=1}^{n-1} \sum_{j=i+1}^n E_i E_j B_{ij} (\cos \tilde{\delta}_{ij} - \cos \tilde{\delta}_{ij}^s) \end{aligned} \quad (8)$$

An approximate way of accounting for TC consists

in adding to the above eq. (7) [respectively (8)] the following term which is not analytically computable and has to be numerically evaluated along the faulted trajectory [6 to 11] :

$$\begin{aligned} I(\delta) &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left\{ M_j \left[\sum_{k \neq i}^n E_i E_k G_{ik} \cos \delta_{ik} \right] \right. \\ &\quad \left. - M_i \left[\sum_{k \neq j}^n E_i E_j G_{jk} \cos \delta_{jk} \right] \right\} d\delta_{ik} \end{aligned} \quad (7')$$

or

$$I(\tilde{\delta}) = - \sum_{i=1}^n \sum_{j=i+1}^n \left\{ E_i E_j G_{ij} \cos \tilde{\delta}_{ij} d(\tilde{\delta}_i + \tilde{\delta}_j) \right\} \quad (8')$$

2.3.2. Individual energy type functions [12,13]

Rather than assessing system's stability through global information contained in the V function, the Iowa group proposes to rely on information contained in an individual function. The leading idea is to attribute the system separation which would cause (if ever) instability, to the motion of a certain "critical" machine, say the i -th, with respect to COA. Thus, the authors of Ref. [12] derive the individual energy type function referred to COA by taking that part of function (8)+(8') which corresponds to this machine

$$\begin{aligned} V_{ic} &= \frac{1}{2} M_i \tilde{\omega}_i^2 - P_i (\tilde{\delta}_i - \tilde{\delta}_i^s) + \frac{M_i}{M} \int_{\tilde{\delta}_i^s}^{\tilde{\delta}_i} P_{COA} d\tilde{\delta}_i \\ &\quad + \sum_{j=1, j \neq i}^n E_i E_j \int_{\tilde{\delta}_i^s}^{\tilde{\delta}_i} [B_{ij} \sin \tilde{\delta}_{ij} + G_{ij} \cos \tilde{\delta}_{ij}] d\tilde{\delta}_i \end{aligned} \quad (9)$$

In Ref. [12] it is claimed that the above function provides less conservative results than the global functions. We believe that its main appeal lies in the reduction of computational burden that it allows. Simulations of Section 5 confirm this conjecture.

If, instead of considering the COA frame we refer to the n -th generator, we derive

$$\begin{aligned} V_{in} &= \frac{1}{2} M_i M_n \omega_{in}^2 - P_{in} (\delta_{in} - \delta_{in}^s) \\ &\quad + M_n E_i \sum_{j=1, j \neq i}^n \left\{ E_j [B_{ij} \sin \delta_{ij} + G_{ij} \cos \delta_{ij}] d\delta_{in} \right. \\ &\quad \left. - M_i E_n \sum_{j=1}^{n-1} \left\{ E_j [B_{nj} \sin \delta_{nj} + G_{nj} \cos \delta_{nj}] d\delta_{in} \right\} \right\} \end{aligned} \quad (10)$$

This latter version of the individual type function is likely to allow further reduction of the computational burden.

Remarks. 1) The success of the individual functions (9) and (10) strongly depends upon the choice of the "critical" machine and, whenever a reference generator is considered, on this reference generator as well. In Ref. [13] a method is provided for identifying the "critical" machine. Section 4 below proposes straightforward identification rules.

2) In the above functions (9) and (10) a good deal of the information is contained in the integral terms which account for TC. This remark has to be kept in mind when looking for combining them with an appropriate practical stability domain estimate.

3) Examination of the pros and cons of the individual vs. global functions is postponed until the end of Section 3.

2.4. TAYLOR SERIES APPROXIMATIONS

2.4.1. Taylor expansion about t_{0+}

An approximate way of expressing $\delta(t_e)$ and $\omega(t_e)$ may be provided by a Taylor series expansion about t_{0+} , i.e. immediately after a disturbance inception (borderline between pre-fault and fault-on periods). This yields the general expressions [14] :

$$\delta_i(t_e) = \sum_{k=0}^n \delta_i^{(k)}(t_{0+}) \frac{(t_e - t_{0+})^k}{k!} \quad (11)$$

where $\delta^{(k)}$ denotes the k -th derivative of δ with respect to time and where the limit conditions are as follows

$$\begin{aligned} \delta_i(t_{0+}) &= \delta_i(t_{0-}) = \delta_i^0; \quad \delta_i^{(2)}(t_{0+}) \equiv \gamma_i = M_i^{-1} [P_{mi} - P_{ei}(t_{0+})] \\ \delta_i^{(2k+1)}(t_{0+}) &= \delta_i^{(2k+1)}(t_{0-}) = 0; \\ \delta_i^{2(k+1)}(t_{0+}) &= -M_i^{-1} P_{ei}^{2(k+1)}(t_{0+}); \quad k = 0, 1, 2, \dots \end{aligned} \quad (12)$$

In Ref [14] the above expansion was used to replace the step-by-step numerical integration from t_{0+} up to CCT. It was shown that for relatively low CCT values the accuracy was very good, and that it was obtained with few terms of the series.

In what follows the Taylor series will be further exploited in two different ways : the one, in next § 2.4.2, for deriving Liapunov-like functions with attractive analytical properties; the other, in Section 4, for identifying unambiguously relevant system machine(s).

2.4.2. Taylor series Liapunov-like functions

Interesting developments of the above series are obtained when truncating eq. (11) after the second term. Indeed, setting $t_{0+} = 0$, eqs. (11) and (12) yield in this case :

$$\delta_i(t_e) = \delta_i^0 + \frac{1}{2} \gamma_i t_e^2; \quad \omega_i(t_e) = \gamma_i t_e \quad (i=1, 2, \dots, n) \quad (13)$$

where γ_i and δ_i^0 are given by (12). From there, we readily derive

$$\delta_{ik} = \delta_{ik}^0 + \frac{\gamma_{ik}}{\gamma_{kl}} (\delta_{kl} - \delta_{kl}^0) \quad (i, k, l = 1, 2, \dots, n) \quad (14)$$

The latter linear approximation makes analytically integrable the terms (7') and (8') as well as those contained in eqs. (9) and (10) which account for TC effects. Moreover, function (9) (respectively (10)) may now be formulated in terms of any single variable (respectively couple of variables). Denoting this latter by i (respectively by in) we get

$$\begin{aligned} V_{icT} &= \tilde{\gamma}_i (\tilde{\delta}_i - \tilde{\delta}_i^0) - [P_i - \frac{M_i}{M} \sum_{j=1}^n P_j (\tilde{\delta}_i - \tilde{\delta}_i^0)] \\ &+ \sum_{j \neq i} \frac{A_{ij}}{\rho_{ij}^i} [\sin(\rho_{ij}^i \tilde{\delta}_i - \beta_{ij}^i) - \sin(\rho_{ij}^i \tilde{\delta}_i^0 - \beta_{ij}^i)] \\ &- \frac{2M_i}{M} \sum_{j=1}^{n-1} \sum_{k=j+1}^n \frac{D_{jk}}{\rho_{jk}^i} [\sin(\rho_{jk}^i \tilde{\delta}_i + \alpha_{jk}^i) - \sin(\rho_{jk}^i \tilde{\delta}_i^0 + \alpha_{jk}^i)] \end{aligned} \quad (15)$$

where $\rho_{jk}^i = \frac{\tilde{\gamma}_{jk}}{\gamma_i}$; $\alpha_{jk}^i = \delta_{jk}^0 - \rho_{jk}^i \tilde{\delta}_i^0$; $\beta_{jk}^i = \theta_{jk} - \alpha_{jk}^i$.

$$\begin{aligned} V_{inT} &= M_i M_n \gamma_{in} (\delta_{in} - \delta_{in}^0) - P_{in} (\delta_{in} - \delta_{in}^0) \\ &+ M_n \sum_{j=1}^n A_{ij} \rho_{ij} \sin(\frac{\delta_{in}}{\rho_{ij}} + \beta_{ij}^0) - \sin(\frac{\delta_{in}^0}{\rho_{ij}} + \beta_{ij}^0) \\ &- M_i \sum_{j=1}^{n-1} A_{nj} \rho_{nj} \sin(\frac{\delta_{in}}{\rho_{nj}} + \beta_{nj}^0) - \sin(\frac{\delta_{in}^0}{\rho_{nj}} + \beta_{nj}^0) \end{aligned} \quad (16)$$

where $\rho_{ij} = \frac{\gamma_{in}}{\gamma_{ij}}$; $\beta_{kj}^0 = \delta_{kj}^0 - \frac{\delta_{kn}^0}{\rho_{kj}} - \theta_{kj}$

Similarly, the global energy type functions may be expressed in terms of a single variable or couple of

variables (see Ref. [15]) :

$$V_T(\tilde{\delta}, \tilde{\omega}) = V_T(\tilde{\delta}_i); \quad V_T(\tilde{\delta}, \omega) = V(\tilde{\delta}_{in}) \quad (17)$$

Remarks. 1) Let us insist on the nice properties obtained from the above truncation of the Taylor series : it leads to a linear dependence of the rotor angle differences; this in turn allows accounting analytically for TCs in either of the above Liapunov functions. Moreover, it allows expressing these functions in terms of a single state variable (δ_i or δ_{in} as appropriate). Further, this makes estimation of the stability domains particularly flexible and straightforward (see § 3.3).

2) In principle, the choice of the i -th variable (respectively in -th pair of variables) is indifferent. In practice, however, it appears interesting to make it correspond to the relevant machine(s) (see Remark 2 of § 4.2).

3) It is obvious that the closer the t_e to the origin (0 or t_{0+}) about which the series is expanded, the better the accuracy of the latter. Stated in terms of disturbance simulations and their corresponding CCTs, the smaller the CCT, i.e. the more severe the disturbance, the more accurate the Taylor series based functions.

4) One could certainly consider the possibility of using a more than 2-term Taylor series. It is however doubtful whether such refinements could enhance accuracy or enlarge the range of CCTs. Indeed, let us first note that this series may approximate only the very first part of the machines' swing curves - and in this case the 2-term truncation is likely to be sufficient. On the other hand, the series, even if it is further expanded, stops being valid as soon as the machines' motion stops conforming to this first part of the "first swing" sine-like shape. We believe, therefore, that the above considered 2-term expansion is an appropriate compromise : it allows analytical simplifications, and it approximates well the actual first swing curves, at least those of the "relevant machines".

2.5. STABILITY DOMAIN ESTIMATES (SDEs)

2.5.1. Theoretical determination of SDE : V_{Th}

For the motion of system (3) in its final configuration, the SDE may *theoretically* be defined upon determining the value that the V function takes on the SDE boundary; in turn, this value may be computed by (see Notation) :

- computing *all* UEPs surrounding the SEP of concern for the system in its final configuration;
- defining V_{Th} as being the value of V on the boundary of the closed hypersurface surrounding SEP :

$$V_{Th} = \min_i V(\tilde{\delta}_{in}^{ui}, 0) \quad (18)$$

The UEP $\tilde{\delta}_{in}^{ui}$ which provides V_{Th} has symbolically been called the "closest" UEP.

2.5.2. Practical computation of $V_{Th} : V_L$

The above computation is unacceptably long for practical purposes because of the very large number of possible UEPs involved : from physical considerations one indeed can propose as many as 2^{n-1} (respect. 2^n) *starting points* (SPs) for solving equations deriving from (3) [respect. (6)] , i.e. equations of the type

$$f_{in}(\tilde{\delta}^e) = 0 \quad [\text{respect. } f_i(\tilde{\delta}^e) = 0]$$

A computationally more convenient procedure consists of :
1° Considering all "type 1" SPs; these are points corresponding to the physical situation where one machine goes unstable with respect to the remaining system. For instance, the i -th type 1 SP is expressed by :

$$\tilde{\delta}_{in}^{si} = [\delta_{1n}^s, \delta_{2n}^s, \dots, \delta_{i-1,n}^s, \pi - \delta_{in}^s, \delta_{i+1,n}^s, \dots, \delta_{n-1,n}^s]^T \quad (19)$$

2° Approximating V_{Th} by

$$V_L = \min_{i=1, 2, \dots, n-1} V(\tilde{\delta}_{in}^{si}, 0) \quad (20)$$

Obviously, the above approximation contributes to speed up significantly computation of SDE. For the conventional stability analysis, which consists of assessing disturbances severity in terms of their corresponding CCTs, this leads to quite a straightforward procedure. Indeed, this latter may be summarized as follows .

- (i) For a given disturbance and system post-fault configuration, compute the approximate limit value, V_ℓ , that V takes on the boundary of its SDE;
- (ii) for successive clearing times, t_e , compute the corresponding values of $V[\delta(t_e), \omega(t_e)]$;
- (iii) define the CCT, t_ℓ , as being the particular t_e for which $V[\delta(t_\ell), \omega(t_\ell)] = V_\ell$.

(21)

However, if the above approximation contributes to substantially reduce the computational burden implied by the theoretical SDE, it does not improve its efficiency with respect to accuracy : the direct criterion still keeps the same erratic and strong conservative character.

Attempts to alleviate this conservativeness have led to the elaboration of three distinct, although similar methodologies [6 to 8; 9 to 11; 16,17]. Their main similarity lies in the fact that all three seek to determine the "interesting" UEP which once reached, it will make the system definitely depart from a possible coming back to synchronism. Notice that this UEP will generally be different from the "closest" one which defines the theoretical SDE; the resulting "Practical" SDE (PSDE) will thus generally be open. An additional conjecture is also common to various methodologies. It states that the "interesting" UEP is related to the machine which, under a given disturbance, would be the first to go out of step, should the disturbance be sustained. This conjecture, in turn, induces a double question : how to identify this machine with certainty (i.e. without risk of failure) and assuming this is possible, how to relate it to the interesting UEP, i.e. how to exploit this information for defining a PSDE. This latter question is answered in the next two paragraphs, where two quite different approaches are described. The reader interested by the justifications underlying these approaches may refer to the indicated references. As for the former question, it will be considered in Section 4.

2.5.3. The Kyoto approach [9 to 11]

N.B. In what follows, V may in principle be any of the functions considered in § 2.3 and 2.4. For the sake of simplicity, however, we will be dealing specifically with global functions; their resulting limit value will be denoted by V_{kg} (k for "Kyoto", g for "global").

By construction, the time derivative of $V(\delta, \omega)$ expressed by (7) is identically zero. On the other hand, at a UEP $V_k(\omega)$ passes by a minimum, zero value. Conjecturing that the trajectory of a sustained disturbance "almost" coincides with the critically cleared one, the authors of this method identify the crossing of the faulted trajectory with the PSDE. Thus, they come up with the following procedure :

- (i) at successive clearing times, t_e , compute the value of the V function, $V[\delta(t_e), \omega(t_e)]$, and of its potential part, $V_p[\delta(t_e)]$, until reaching a maximum of the latter. This will be considered to be the limit value, V_{kg} , taken by V on the boundary of the PSDE :

$$V_{kg} = V_p \max \quad (22)$$

- (ii) the CCT, t_{kg} , will accordingly be found at the intersection of $V[\delta(t_e), \omega(t_e)]$ with V_{kg} :

$$V[\delta(t_{kg}), \omega(t_{kg})] = V_{kg} \quad (23)$$

2.5.4. The acceleration approach [16,17]

The underlying conjecture is that the interesting UEP is related to the machine which the first would go

out of step should the considered disturbance be sustained. Moreover, this UEP is assumed to be very close to its corresponding starting point (SP) of form (19), so that the value of the V function at SP is a good approximation of its value at UEP. Thus, the question of concern becomes again how to identify the relevant machine. In Ref. [17], this machine was conjectured to be the one possessing the maximum *initial acceleration*, i.e. the maximum acceleration at t_{0+} ; moreover, the reference machine was chosen to be the one with the minimum initial acceleration. In Section 4, refined rules are used to identify what we shall symbolically call the "fastest" and "slowest" machines respectively.

The acceleration approach consists of the following procedure .

- (i) At t_{0+} , i.e. immediately after the considered disturbance inception, determine the "fastest" and "slowest" machines. Denoting them respectively by i and n , define the "interesting" UEP to be given by (19) and compute accordingly the limit value V_a :

$$V_a = V(\delta_{i1}, 0) \quad (24)$$

- (ii) At successive clearing times, t_e , compute the value of V , $V[\delta(t_e), \omega(t_e)]$, until reaching V_a ; the CCT, t_a , is given by

$$V[\delta(t_a), \omega(t_a)] = V_a \quad (25)$$

Remark. When a cluster of machines, instead of a single one, are shown to be (almost) equally "fast", the interesting UEP of concern is no longer the SP of form (19), of "type 1", but rather a "type higher than 1" appropriately expressed.

2.5.5. Kyoto vs. acceleration approach

1. Because of the very way of determining the limit value of V in the Kyoto approach, evaluation of V has to be pursued from t_{0+} up to $V_{pmax} = V_{kg}$. This is an important advantage over the acceleration approach and a rather minor drawback. The advantage lies in the fact that since V_{kg} is path dependent, accounting for TCs by integrating along the path becomes quite easy in either of the global (term (7')) or (8')) and the individual (integral terms in (9) and (10)) functions. On the contrary, the acceleration approach cannot take TCs into account (how would it be possible while V_a is path independent?). Now, this advantage is somewhat counterbalanced by a longer computing time necessitated for the Kyoto approach : our experience suggests that V_{pmax} is reached after 20 % to 50 % above CCT; for large power systems, the corresponding additional computing time may be significant, whereas for small ones it is not.

2. A theoretical limitation of the Kyoto approach is linked to its basic assumption $\dot{V}=0$. Hence, this approach cannot take into account damping effects which would make $\dot{V}$ negative (semi-)definite. This however has no important practical consequences, at least for the time being [3].

3. A more important drawback of the Kyoto approach may - and does - exist. It is linked to the very method's principle, which consists in declaring that the limit value of V is reached as soon as a maximum of $V_p(\delta)$ is met. But such a maximum may be a local, unimportant one. In this case the Kyoto approach will lead to unduly conservative results. Such cases are observed when simulating relatively mild disturbances (see § 5.3 below).

4. With respect to computational requirements, both the Kyoto and the acceleration approaches are simple and fast. Indeed, the determination of the CCT corresponding to a given disturbance requires the following computations :

- (i) running a load flow program;
- (ii) running a step-by-step stability program for a short duration, (see (v) below);
- (iii) computing the admittance matrix reduced to the generator internal nodes of the system in its final

configuration;
 (iv) computing the SEP of the system in its final configuration;
 (v) for the acceleration approach : computing V_a and $V[\delta(t_e), \omega(t_e)]$ from t_{o+} up to t_a ; for the Kyoto approach : computing $V[\delta(t_e), \omega(t_e)]$ and $V_p[\delta(t_e)]$ from t_{o+} up to reaching a maximum of $V_p[\delta(t_e)]$, i.e. up to a t_e larger than t_{kg} (see above).

3. DEVISING PRACTICAL DIRECT CRITERIA

Many variants may be devised from the combination of functions cited in § 2.3 and 2.4 with PSDEs cited in § 2.5.3, 2.5.4. In what follows we shall quote the most interesting among them, i.e. the approaches which are more likely to exhibit good performances with respect to reliability, accuracy, and computational burden and speed. We shall group them in three broad classes. The first two collect combinations of the Kyoto and the acceleration approaches with the energy type functions of § 2.3. The third group gathers all the Taylor series based criteria, combination of PSDEs of § 2.5.3 and 2.5.4 with functions of § 2.4.

3.1. KYOTO BASED CRITERIA

3.1.1. Kyoto global criteria

Deriving from the combination of the global function (7), (7') with the limit value of § 2.5.3, they yield the following two global criteria :
Kyoto global "without" TCs, combining V function (7) with limit value of form (23);
Kyoto global "with" TCs, combining function (7)+(7') with limit value of form (23).

Note that two other similar combinations are provided when considering functions (8), or (8)+(8'); they have equivalent performances with the corresponding above ones with respect to reliability and accuracy but they require more important computational burden.

3.1.2. Kyoto individual criteria

We shall distinguish the following two :
Kyoto individual in the COA reference frame, combining function V_{ic} (9) with limit value of form (23);
Kyoto individual in the n-th generator's reference frame, which combines function V_{in} (10) with limit value of form (23).

3.2. ACCELERATION BASED CRITERIA

The acceleration based PSDE being unable to account for TCs (see § 2.5.5.1), individual functions appear inappropriate, at least as long as they contain non-analytically integral TC terms (see Remark 2 in § 2.3.2).

In what follows, we shall consider the combination of global function (7) with the limit value (25).

3.3. TAYLOR SERIES BASED CRITERIA

The interesting analytical properties of functions of § 2.4 open out into a variety of criteria. We shall merely quote few of them and finally select two among the most promising ones.

3.3.1. Taylor series-Kyoto based criteria

Either of the global (17) and the individual [V_{ict} (15), V_{int} (16)] functions may be used, and for each of these many variants may be thought of. E.g., such variants may derive through the procedure suggested in [18]; denoting by $V = V_k(\omega) + V_p(\delta)$ any of the above-mentioned functions, we summarize this procedure as follows :

- 1° express the derivative $\frac{\partial V_p}{\partial \delta_{in}} \bigg|_{\delta_{in}^u} = 0$ and compute the corresponding δ_{in}^u ;
- 2° compute $V_{pmax} = V(\delta_{in}^u)$;
- 3° let τ_k denote the CCT provided by this method; then

$$V[\delta_{in}(\tau_k)] = V_{pmax} \quad (26)$$

$$\text{further, let } \delta_{in}^k = \delta_{in}(\tau_k) ; \quad (27)$$

4° solve eq. (13) in terms of δ_{in}^k to compute the corresponding CCT, τ_k :

$$\tau_k = \{2(\delta_{in}^k - \delta_{in}^0) / \gamma_{in}\}^{1/2} \quad (28)$$

Note that the time τ_k may be obtained either through the Taylor series (13) used to compute $\delta_{in}(t_e)$ for successive t_e values, or by solving the non-linear eq. (26) with respect to δ_{in}^k .

3.3.2. Taylor series-acceleration based criteria

The acceleration approach becomes more accurate as soon as it becomes able to take TCs into account. In such a case, it becomes also able to adapt to individual type functions.

The procedures reduce then to the following steps :

- (i) compute V_a of form (25), where δ_{in}^u reduces to $\pi - \delta_{in}^s$;
- (ii) adapt above step 3°;
- (iii) adapt above step 4°.

Note that this kind of criteria lead back to the well-known "equal area" type.

3.4. COMPARING THE VARIOUS DIRECT CRITERIA

3.4.1. Kyoto based criteria

The individual criteria compared to the global ones reveal the following :

cons : the former need relevant machines' identification whereas the latter don't;

pros : - the former require much less computing efforts and time, especially with the V_{in} function, and especially when they are compared (as they should be) with the global "with" TCs;

- in some few cases they show more accurate.

On the whole, the pros prevail; this suggests that the Kyoto based individual criteria should be preferred to the global ones.

3.4.2. Acceleration based criteria

The computing efforts and time are less important for the acceleration than for the Kyoto based criteria, even the Kyoto individual ones. But the former imply neglect of TCs and hence are likely to be less accurate. On the other hand, however, they have a wider domain of validity and show reliable under all circumstances (see Remark 3 of § 2.5.5 and simulation results of § 5.3).

3.4.3. Taylor series based criteria

Within their (rather) limited range of validity they exhibit the best performances with respect to both accuracy and computing efforts and time. Among them, the fastest and simplest one is that of § 3.3.2.

3.4.4. Overall direct criteria's assessment

We tentatively propose the following classification, based on a decreased severity of the stability conditions that the direct criteria are called to assess. This classification results from the above theoretical-physical-pragmatic considerations; it is corroborated by the simulation results of Section 5.

For very stringent stability conditions : without contest, the Taylor series based criteria have the best performances.

For rather stringent stability conditions : the Kyoto individual criteria prevail.

For all kinds of stability conditions : the acceleration (global) criteria reveal always reliable.

4. RELEVANT MACHINES IDENTIFICATION

In the purely theoretical form of Liapunov's direct method, there are no "privileged" among the system machines : any of them may interchangeably act as the reference, and no information about the machine which the

first is going to become unstable (if ever) is needed nor provided. However, in most of (and in fact in all the interesting among) the previously devised direct criteria, knowledge of these relevant machines is of paramount importance. In fact, depending upon the criterion used, the following one or two machines are to be identified: (i) the machine which the first would cause the loss of system's synchronism if the disturbance was sustained longer than CCT; i.e. the machine whose rotor angle with respect to a reference increases, during the fault-on period, beyond a certain limit value (e.g. beyond 180°); this is what we shall symbolically call the "fastest" machine; (ii) the other machine of concern is the reference generator, whenever it is used; we shall symbolically call it the "slowest" machine and make it correspond to that whose rotor angle time evolution shows to be the mildest one.

4.1. MACHINES CLASSIFICATION

To identify the relevant machines of a system in a given configuration and for a specified disturbance, we shall take advantage of the information provided by the machines behaviour at t_{0+} , i.e. by the values of their initial rotor angles, accelerations and second derivatives of the latter. This information derives readily from eqs. (11), (12):

$$\begin{aligned} \delta_i(t_{0+}) &= \delta_i(t_{0-}) = \delta_i^0; \quad \gamma_i = M_i^{-1} [P_{mi} - P_{ei}(t_{0+})] \\ &\quad (i=1, 2, \dots, n) \quad (29) \\ \dot{\gamma}_i &= 0; \quad \ddot{\gamma}_i = M_i^{-1} \sum_{j=1}^n \gamma_{ij} A_{ij} \sin(\delta_{ij}^0 - \theta_{ij}) \\ &\quad j \neq i \end{aligned}$$

Note that if the considered disturbance is a three-phase short circuit acting at a machine's (say the k -th's) bus-bar, eqs. (29) reduce for $i=k$ to

$$\gamma_k = M_k^{-1} P_{mk}; \quad \ddot{\gamma}_k = 0 \quad (30)$$

4.2. IDENTIFICATION RULES

(i) Classify the machines in a decreasing order of their γ ; the candidate "fastest" machines will be among those possessing significantly larger (positive) values of γ , the candidate "slowest" ones among those possessing the smallest values of γ (they may be close to zero or negative).

(ii) Explore the magnitude and sign of $\ddot{\gamma}$ for the above candidate machines. The fastest one will be that which combines larger γ and positive $\ddot{\gamma}$; the slowest will be that which combines smaller γ and negative $\ddot{\gamma}$.

(iii) In case the above tests show unable to clearly identify the relevant fastest machine, consider the candidate fastest ones, compute their corresponding $\delta_i(t_e)$ according to

$$\delta_i(t_e) = \delta_i^0 + \frac{1}{2} \gamma_i t_e^2 + \frac{1}{24} \ddot{\gamma}_i t_e^4; \quad \omega_i(t_e) = \gamma_i t_e + \frac{1}{6} \ddot{\gamma}_i t_e^3 \\ (t_e = 0, 0.10, 0.20, 0.30) \quad (31)$$

and declare fastest the one with larger (δ, ω) at $t_e = 0.20$ or 0.30 s.

(iv) In doubtful cases, i.e. when many machines show still "equally fast" proceed as follows:

- for the acceleration approach: consider the cluster of the above machines;
- for the individual functions: use successively each of the above machines and consider the smallest among the critical clearing times to be the actual one.

Note that according to our experience the latter computation has to be repeated no more than two times (see Section 5).

(v) In case tests (i) and (ii) show unable to identify clearly the "slowest" machine, any one of the serious candidates can be used; indeed, experience shows that the results will hardly be affected when interchanging reference generator, as long as it is taken among the above candidates.

Remarks. 1) The above identification rules imply extremely simple logic and hence extremely simple and

straightforward computations.

2) It is reasonable to think that the "faster" the machine considered, the better the accuracy of series (31). Indeed, this series is valid for small (<1) t_e values for which terms in $t_e^4, t_e^6, \dots$ become negligible. Therefore, only machines with sine-like swing curves will be accurately described; hence, the more accurate Taylor series expansions correspond to the faster machines, among which is the relevant machine. This conjecture is corroborated by our numerical simulations.

3) The Taylor series based criteria have to obey the γ -test solely, since they derive from a 2-term Taylor expansion.

5. SIMULATIONS

5.1. SIMULATION DESCRIPTION

5.1.1. The systems

Eight systems have been investigated comprising real (or reduced versions of) power networks along with some test-systems so as to cover a wide range of system characteristics. They are identified as follows.

- * 3-machine test system
- * 6-machine Tunisian EHV simplified power system
- * 7-machine CIGRE system
- * 9-machine power system
- * 14-machine Greek EHV simplified power system (N.B. The configuration used here corresponds to its situation in the early seventies)
- * 15-machine test system
- * 17-machine Iowa system, reduced version of the network of the State of Iowa
- * 40-machine Belgian EHV simplified power system.

5.1.2. The disturbances

The first group of disturbances is of the standard three-phase short circuit (3ØSC) type. Their location concerns almost always Generators' Bus-bars (GB), apart from a few exceptions relative to Other than generators' Bus-bars (OB).

A second group of disturbances consists of line trippings and load sheddings.

All disturbances are supposed to be finally cleared, i.e. that the system's post-fault configuration is identical to the pre-fault one.

5.1.3. The results

N.B. The results listed in this paper are only a representative sample of those actually obtained. Ref. [15] give a more detailed and extensive account of the overall set of our simulations.

Results relative to 3ØSC simulations are commented in § 5.2 and summarized in Tables I and II. Those relative to other types of disturbances are reported in § 5.3.

The standard step-by-step method is used as the benchmark in our comparisons. Its CCTs have been assessed through numerical integration of the generators' swing curves pursued up to 1.50 s., except when instability arises earlier. All CCTs are expressed in seconds. They correspond to the stable (lower bound) case.

A dozen of practical methods or variants have been investigated, eight of which appear in the Tables hereafter.

The simulations sought to explore reliability, accuracy, and computational efficiency. The reliability and the accuracy of a method are assessed through comparisons of CCTs provided by it and by the step-by-step one, under the same simulation conditions and, whenever possible, with the same system modelling. Evaluating computational efficiency is somewhat more hazardous; here, an approximate first estimation is only given.

5.2. RESULTS OBTAINED WITH 3ØSC TYPE DISTURBANCES

5.2.1. Organization

Table I. It collects information relative to reli-

TABLE I

	1	2	3	4	5	6	7	8	9	10	11	12
Machine	Fault locat. Bus #	CCT s.b.s. t_b	"Fastest" gener.	Refer. gener. (n)	Conven. criterion t_g	Accel. criterion t_a	Kyoto global (without) t_{kg}	Kyoto global (with) t_{kg}	Kyoto individ. ref. gen. t_{kin}	Kyoto individ. COA t_{kic}	Taylor Kyoto individ. r_{kin}	Taylor Accel. individ. r_{ain}
3-machine	1 2 3	0.54 0.37 0.32	1 2 3	2 1 2	0.47 0.40 0.27	0.47 0.40 0.33	0.72 0.40 0.33	0.72 0.37 0.32	0.55 0.37 0.33	0.56 0.38 0.32	0.51 0.39 0.32	0.48 0.38 0.32
6-machine Tunisian	1 2 3 4 5 6	0.30 0.86 0.38 > 1.50 0.26 0.39	1 3 3 4 5 6	4 4 4 2 4 4	0.36 0.51 0.32 1.07 0.17 0.25	0.36 0.98 0.41 > 1.50 0.28 0.41	0.37 0.98 0.42 1.42 0.28 0.42	0.30 0.92 0.38 1.34 0.27 0.39	0.30 0.90 0.38 1.35 0.25 0.39	0.30 0.91 0.38 1.35 0.27 0.39	0.31 0.48 0.39 0.80 0.25 0.38	0.31 0.49 0.39 0.88 0.25 0.33
7-machine CIGRE	1 2 3 4 5 6 7	0.37 0.41 0.39 0.52 0.35 0.52 0.33	1 2 3 4 5 6 7	2 7 2 2 2 2 2	0.32 0.41 0.32 0.37 0.33 0.43 0.45	0.38 0.42 0.39 0.54 0.38 0.50 0.34	0.37 0.41 0.39 0.54 0.35 0.50 0.33	0.38 0.41 0.39 0.51 0.35 0.52 0.33	0.38 0.41 0.39 0.52 0.35 0.51 0.33	0.38 0.41 0.39 0.52 0.35 0.52 0.33	0.40 0.41 0.40 0.52 0.38 0.58 0.34	0.36 0.41 0.37 0.45 0.35 0.48 0.34
9-machine	1 2 3 4 5 6 7 8 9 15 (OB) 24 (OB) 26 (OB)	0.45 0.44 0.27 0.60 0.50 0.48 0.60 0.50 0.48 0.47 0.74 0.36	1 2 3 4 5 6 7 8 9 6 7 3	9 8 8 9 9 9 5 8 9 8 8	0.48 0.31 0.22 0.52 0.50 0.34 0.45 0.42 0.48 0.33 0.54 0.28	0.48 0.46 0.28 0.64 0.55 0.51 0.65 0.51 0.51 0.56 0.78 0.36	0.49 0.46 0.28 0.65 0.55 0.51 0.65 0.53 0.52 0.56 0.80 0.40	0.45 0.44 0.27 0.60 0.50 0.48 0.60 0.50 0.49 0.54 0.74 0.38	0.45 0.44 0.27 0.61 0.50 0.48 0.60 0.50 0.48 0.50 0.76 0.38	0.45 0.44 0.27 0.61 0.50 0.48 0.61 0.50 0.49 0.50 0.75 0.38	0.48 0.45 0.27 0.54 0.54 0.47 0.63 0.50 0.49 0.52 0.67 0.38	0.43 0.43 0.27 0.53 0.53 0.45 0.57 0.49 0.47 0.53 0.57 0.36
14-machine Greek	17 19 21 33 34 59 76 77 84 89	0.58 0.52 0.50 0.64 0.32 0.48 0.47 0.47 0.26 0.30	17 19 21 33 34 59 76 77 84 89	21 21 59 17 17 21 64 76 0 76	0.29 0.26 0.17 0.22 0.08 0.19 0.17 0.26 0.07 0.09	0.65 0.53 0.50 0.67 0.35 0.49 0.49 0.47 0.27 0.30	0.65 0.55 0.50 0.69 0.36 0.51 0.49 0.49 0.27 0.31	0.62 0.52 0.49 0.65 0.33 0.48 0.48 0.48 0.26 0.30	0.59 0.52 0.49 0.65 0.31 0.48 0.48 0.50 0.26 0.31	0.57 0.51 0.49 0.65 0.33 0.48 0.48 0.48 0.26 0.30	0.51 0.47 0.48 0.65 0.32 0.47 0.50 0.47 0.26 0.32	0.48 0.45 0.46 0.58 0.31 0.47 0.47 0.49 0.25 0.30
15-machine	M1 K1 Q1 O1 C1 G1 N1 ZLU	0.36 0.36 0.43 0.40 0.42 0.42 0.40 0.51	M1 K1 Q1 O1 C1 G1 N1 ZLU	M2 M2 G2 G2 G2 G2 K2 M1	0.34 0.37 0.28 0.28 0.38 0.38 0.39 0.37	0.38 0.38 0.45 0.45 0.46 0.46 0.39 0.55	0.39 0.49 0.47 0.39 0.47 0.47 0.50 0.55	0.36 0.36 0.44 0.43 0.42 0.42 0.44 0.51	0.36 0.36 0.43 0.43 0.42 0.43 0.42 0.52	0.36 0.36 0.42 0.41 0.42 0.42 0.42 0.51	0.37 0.37 0.45 0.45 0.46 0.46 0.42 0.54	0.37 0.37 0.40 0.40 0.43 0.43 0.42 0.51
17-machine Iowa	G2 G3 G4 G5 G10 G14 G17 COOPER RAUN CALHOUN	0.21 0.27 0.36 0.23 0.24 0.40 0.22 0.22 0.20 0.33	G2 G3 G4 G5 G10 G14 G17 G2 G6 G16	G1 G14 G14 G14 G14 G13 G14 G1 G15 G14	0.14 0.13 0.12 0.15 0.23 0.14 0.13 0.11 0.10 0.14	0.24 0.30 0.39 0.25 0.27 0.41 0.23 0.20 0.22 0.26	0.25 0.30 0.39 0.27 0.28 0.41 0.24 0.28 0.22 0.44	0.21 0.27 0.38 0.23 0.23 0.24 0.39 0.21 0.22 0.19 0.36	0.23 0.26 0.37 0.23 0.23 0.24 0.41 0.21 0.22 0.21 0.34	0.21 0.27 0.38 0.23 0.23 0.24 0.39 0.21 0.21 0.21 0.34	0.23 0.27 0.39 0.23 0.23 0.24 0.41 0.21 0.21 0.21 0.25	0.23 0.27 0.36 0.23 0.23 0.25 0.41 0.21 0.21 0.20 0.25
40-machine Belgian	AWIR3 AWIR4 BRES4 VILG4 DOEL1B MERC1(OB) FARC4 LANB3 COO1 GRAM1	0.24 0.23 0.26 0.25 0.25 0.25 0.27 0.28 0.33 0.33	AWIR3 AWIR4 BRES4 VILG4 DOEL1B DOEL1A FARC4 RODH3A COO1 GRAM1	MAAS1 REV11 REV11 MAAS1 ROMK1 MAAS1 ROMK1 ROMK1 ROMK1 ROMK1	0.11 0.16 0.17 0.20 0.11 0.09 0.24 0.14 0.17 0.10	0.24 0.27 0.29 0.29 0.24 0.27 0.32 0.27 0.33 0.33	0.27 0.28 0.30 0.30 0.31 0.31 0.33 0.33 0.37 0.38	0.23 0.23 0.25 0.26 0.25 0.24 0.27 0.27 0.34 0.33	0.24 0.23 0.26 0.26 0.25 0.25 0.27 0.27 0.34 0.33	0.24 0.24 0.26 0.26 0.25 0.25 0.27 0.27 0.34 0.33	0.24 0.24 0.27 0.26 0.25 0.25 0.27 0.27 0.34 0.33	0.24 0.24 0.27 0.26 0.25 0.25 0.27 0.27 0.34 0.33

TABLE II

1	2	3	4			5			6			7	8	9	10	11	12
SYSTEM DATA			METHOD ACCURACY									CPU COMPUTING TIMES (T) . RATIOS IN %					
n	N	B	GLOBAL			INDIVIDUAL 1			INDIVIDUAL 2			s.b.s. (T _b)	T _G T _b	T ₁₁ T _G	T ₁₂ T _G	T ₁₁ T _G	T _r T _r +T ₁₁
			I	II	III	I	II	III	I	II	III						
6	20	26	> .16	4/5	4/5	> .14	4/5	4/5	> .15	4/5	4/5	15	11	60	12	8	9
7	10	13	.01	7/7	7/7	.01	7/7	7/7	.01	7/7	7/7	9	14	66	63	4	3
9	34	36	.07	11/12	10/12	.03	11/12	9/12	.03	11/12	10/12	18	9	54	76	5	8
14	92	119	.04	9/10	9/10	.03	9/10	9/10	.01	10/10	10/10	40	14	41	70	1.7	8
15	44	99	.04	6/8	6/8	.03	7/8	6/8	.02	8/8	7/8	32	22	41	57	1.4	9
17	162	281	.03	9/10	8/10	.02	10/10	9/10	.02	10/10	9/10	147	8	37	72	0.2	64
40	163	269	.01	10/10	10/10	.01	10/10	10/10	.01	10/10	10/10	100	95	19	31	0.8	37

ability and accuracy aspects. It is organized in the columns numbered 1 to 12 as follows.

- #1: disturbance location : unless otherwise specified it concerns GBs;
- #2: CCT provided by the step-by-step method : t_b ;
- #3: identification of the relevant "fastest" machine;
- #4: identification of the relevant "slowest" machine;
- #5: CCT provided by the conventional Liapunov criterion : t_l ;
- #6: CCT provided by the acceleration criterion of § 3.2; the considered relevant machines are those of columns 3 and 4, except for cases where "other than type 1" UEPs are considered; these cases are indicated by ♦ : t_a ;
- #7: CCT provided by the Kyoto "global" criterion of § 3.1.1, "without TC" : t'_{kg} ;
- #8: CCT provided by the Kyoto "global" criterion of § 3.1.1, "with TC" : t_{kg} ;
- #9: CCT provided by the Kyoto individual criterion of § 3.1.2 : t_{kin} ;
- #10: CCT provided by the Kyoto individual criterion of § 3.1.2 : t_{kic} ;
- #11: CCT provided by the Taylor series based criterion of § 3.3.1 : τ_{kin} ;
- #12: CCT provided by the Taylor series based criterion of § 3.3.2 : τ_{ain} .

Table II. It assembles three groups of information.

* The first concerns investigated systems' essential topological data :

- #1: number of system generators (n);
- #2: total number of system buses (load plus generator nodes) (N);
- #3: total number of system branches (B).

* The second group summarizes the information of Table I. The compared methods are reported in columns 4 to 6 as follows :

- #4: accuracy of the Kyoto "Global" method, listed in #8 of Table I;
- #5: accuracy of the Kyoto "Individual 1", listed in #9 of Table I;
- #6: accuracy of the Kyoto "Individual 2", listed in #10 of Table I.

Moreover, each of the above three columns is subdivided in three sub-columns providing the following :

Sub-column I : maximum discrepancy of method's CCT with respect to t_b , in s.;

Sub-column II : number of simulations yielding CCTs within $t_b \pm 0.02$ limits over total number of simulated 3ØSCs;

Sub-column III : number of simulations yielding CCTs within $t_b \pm 0.01$ limits over total number of simulated 3ØSCs.

* The third group of Table II gives a first approximate evaluation of the methods' computing times in % of CPU :

- #7: compares the computing time required by the step-by-step method for evaluating t_b ; three trials have been assumed necessary : $t_{e1} < t_b$, $t_{e2} = t_b$, $t_{e3} > t_b$; the t_b considered here are identified in italics in Table I;
- #8: compares approximately the ratios in % of the computing times needed respectively by the "Global" (T_G) and by the step-by-step (T_b) methods;
- #9,10,11: compare the computing times needed respectively by the "Individual 1" approach (T_{I1}) by the "Individual 2" (T_{I2}), and by the "Taylor Individual 1" approach (T_{TI1}), all with respect to T_G ; the ratios are expressed in %; note also that these comparisons do not take into account the computing time needed for the computation of the admittance matrix reduced at the generators' nodes;
- #12: ratio of the CPU time needed to compute Y_r (T_r) with respect to the overall CPU time needed by the "Individual 1" approach.

5.2.2. Discussion

5.2.2.1. Relevant machines' identification. It has con-

formed to the rules of § 4.2. A first selection was accordingly made on the basis of the magnitude and sign of the machines initial accelerations; when necessary, $\ddot{\gamma}$ -test was used to decide between (almost) equi- γ machines; the Taylor series (29) contributed to further clarify some few ambiguous cases.

As concerning the "fastest" machines, observe that generally a 3ØSC acting at the GB of a machine confers to this latter the largest γ (and also $\dot{\gamma}=0$). The exceptions encountered have been satisfactorily clarified through the successive refined tests. In the very few cases where an ambiguity still exists, we have considered a "type higher than 1" UEP for the acceleration criteria; as for the individual type criteria we have repeated the test as many times as necessary (2 times at the maximum). Even the OB cases which are generally less obvious, have thus been treated unambiguously.

With regard to the "slowest" machines, their identification have sometimes been less obvious (see Remark 2 of § 4.2) but also less crucial: all "candidate slowest" machines provide very close - if not identical - results.

5.2.2.2. Sample of peculiar cases. * A typical peculiar example is the 3ØSC at GB #1 of the 3-machine system [15]. In this case we observe that relying solely on the γ and $\dot{\gamma}$ values would have led to declare "fastest" the machine 1 which in turn would have provided $t'_{kin}=0.70$ s., $t'_{kic}=0.72$ s., i.e. unduly large CCTs. Exploration of the machines angles evolution according to eq. (29) suggests that machines 1 and 3 are almost "equi-fast"; this leads to consider a "type 2" UEP for the acceleration criterion and machine #3 as the relevant machine for the individual criteria. This yields accurate results for the acceleration and the individual criteria; on the contrary, the Kyoto global fails.

* Another peculiar case concerns the 3ØSC at GB #4 of the Tunisian system, which appears to correspond to a very mild disturbance ($t_b > 1.50$ s.). Except for the acceleration criteria, all others show here unreliable; this case will be imbedded in the discussions of § 5.3.

5.2.2.3. Accuracy evaluation

* *Conventional direct criterion* (column #5 of Table I). Notice the erratic results of this column.

* *Influence of TCs* (columns #6,7,8 of Table I). The results are generally in good agreement with t_b . Including TC allows further improvements, as it may be seen by comparing columns #7 and 8 relative to the Kyoto global criteria. The same observation holds for the acceleration criteria according to simulations not reported here : when the loads are localized at GBs, the approximation of TC neglect is improved, and so is the accuracy of t_a .

* *Global vs. individual (v_{in}) vs. individual (v_{ic}) functions* (Columns #8,9,10 of Table I). The three methods behave almost equally well, with, however, a slight advantage of the two latter over the former (see also columns #4 to 6 of Table II).

* *Taylor's approximation.* For a limited range of CCTs, say for CCTs not exceeding 0.45 s., both τ_{kin} and τ_{ain} are accurate. Other Taylor approximations are investigated in [15]. It is found that the 2-term expansion based ones yield results very similar, if not identical, to those reported in Table I. The higher order expansions, on the other hand do not bring any benefit with regards to accuracy, while losing the advantages of the 2-term expansion.

5.2.2.4. Estimating computational efficiency (Table II)

N.B. During our simulations, computational efficiency was not sought. Therefore, the Computing Times (CT) can only give a crude indication; this is why we proceed by comparisons. To make them valid, we have used simulations implying similar CCTs in the various systems. Indeed, the rapidity of the step-by-step integration is different in the fault-on and the post-fault periods. On the other hand, we have split the proced-

ures into two parts and estimated their corresponding CCTs separately; the one concerns the reduction of the admittance matrix, the other the rest of the procedure. Note that for 30SC at GBs the reduced admittance matrix of the fault-on period may be derived from that of the pre-fault one through the superposition technique suggested in [3]; thus, for this type of disturbances, matrix reduction is computed once for all.

* Column #7. The T_b ratios generally increase with the system size. The 100 % choice is obviously arbitrary.

* Column #8. The gain in CT increases rapidly with the system size.

* Columns #9,10. Although similar, T_{11} is always smaller than T_{12} .

* Column #11. Notice the substantial gain in CT for medium-sized and even more for large systems.

5.3. OTHER TYPES OF DISTURBANCES

The second group of disturbances concerns loss of generators, Line Trippings (LT), and Load Sheddings (LS). Table III lists some of our investigation results. Obviously, all direct criteria show here unreliable, except for the acceleration one. The explanation for these failures and for that of the Tunisian system may be found in Remark 3 of § 2.5.5 and in § 3.4.2. An illustration of the phenomena is given in Ref. [15].

Remark. One could maybe argue that disturbances which correspond to CCTs exceeding certain values (say > 1 s.) are beyond the scope of transient stability analysis and of direct criteria. Nevertheless, a reliable method should be required to never yield such misleading results as in Table III; by no means should it declare unstable a system which actually is very far from instability.

TABLE III

Disturb. identifi.	t_b	t_l	t_a	t'_{kg}	t_{kg}	t_{kin}	t_{kic}
1	0.32	0.08	0.35	0.35	0.32	0.33	0.33
2	>1	>1	>1	0.38	0.38	0.66	>1
3	>1	>1	>1	0.33	0.33	0.51	>1
4	>1	>1	>1	0.28	0.28	0.35	0.35
5	>1	>1	>1	0.44	0.43	0.42	0.37

Dist #1 : 14-machine Greek system : LT of the double line 32-72
 Dist #2 : 14-machine Greek system : LS of the load 191.7 MW at GB #1
 Dist #3 : 14-machine Greek system : LS of the load 344.3 MW at OB #30
 Dist #4 : 40-machine Belgian system : LS of the load 267 MW at GB #MOL3
 Dist #5 : 40-machine Belgian system : LS of the load 855 MW at OB #AVEL1

5.4. SELECTING APPROPRIATE CRITERIA

Scanning the foregoing, one sees emerge three different classes of direct criteria, appropriate to cover different stability assessment needs. But how to choose a priori among them, for instance during the real-time operation of a power system ? An interesting hint is provided by the observation that the range of the accelerations of a system machines is typically related to the severity of the stability conditions. Ref. [15] discusses this question and reports numerical examples illustrating the above observation.

6. CONCLUSIONS

Many energy Liapunov-like direct criteria have been devised, classified and investigated. Their merits and limitations have been assessed and compared. Three classes of methods have thus emerged.

The first comprises the Taylor series based criteria. Within their narrow range of validity (which covers only quite stringent stability conditions) they are very accurate and extremely fast. The optimal among these criteria seems to be the "Taylor-acceleration-individual" one.

The second class groups the Kyoto based criteria. Of them, we select the individual ones, the most efficient of which being the "Kyoto-individual-reference generator" criterion. Their validity covers very strin-

gent up to stringent stability conditions. Within these limits they are very accurate and fast.

The third class consists of the "acceleration-global" criterion. It is valid in the whole field of transient stability; although not so accurate as the previous ones, it is the most reliable and may be used as a "safe-guard" procedure, inasmuch as it is very fast.

The accuracy and reliability of all of the above methods are not influenced by the system size; their computational performances, on the other hand, vary quite smoothly with it.

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